PhD Workshop Course Series

Title: Processing Data using Python

Objective: The session aims to provide a foundational understanding of common data manipulation techniques essential for preparing data for analysis, methods to verify your own work, and to address common challenges when working with financial statement panel data. No prior programming experience is required.

Target Audience: Ideal for participants looking to design and implement research workflows, from raw source data to analysis-ready datasets. Examples include financial statement panel data and event-shaped datasets.

After this workshop, you will be able to:

- Set up a local Python virtual environment for data analysis using VS Code, Miniforge, and Jupyter Notebooks.
- Import, clean, and transform datasets using Pandas and DuckDB.
- Perform joins, unions, aggregations, and filtering operations on data frames.
- Identify and handle missing values and duplicates.
- Connect to WRDS and query data.
- Learn basic SQL syntax.
- Structure your code for reproducibility and efficiency.
- Store data efficiently using Parquet format.
- Be aware of RAM constraints when working with data.
- Validate and verify your results.
- Parallelization

Topics:

- Overview
    - Tools and data structures
        - DataFrames (Pandas)
    - Code workflow
    - Jupyter Notebook
        - Browsing the data
- Joins (Horizontal merges)
    - Inner, left, right, outer, and cross
- Unions (Vertical merges)
- Introduction to SQL
    - Pulling data from WRDS using SQL (PostgreSQL)
    - DuckDB
- Saving your work

    - Code
    - Data
- Aggregating data (row aggregation by groups)
    - Sum, mean, median, std, concat
- Dealing with duplicates
- Lags and leads
    - Challenges of lagging and leading when there are gaps or duplicates
- Linking tables and ranged joins
- Parallelization